

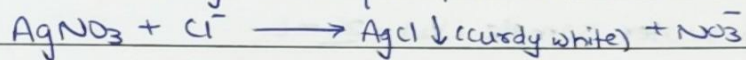
◦ Aim:

- To estimate the chloride content of a given sample of water, being provided with 0.05N silver nitrate solution.

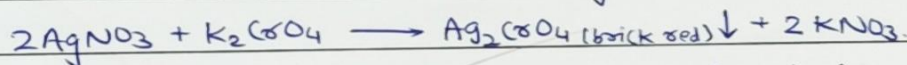
◦ Theory:

- Chloride content of water is determined by titrating a known volume of water with silver nitrate using potassium chromate as indicator.

- Silver nitrate reacts quantitatively with chloride ions present in the water to form a curdy white precipitate of silver chloride.

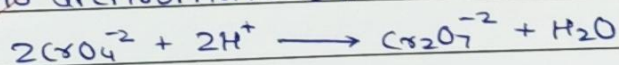


- When chromate ions of the indicators are also present in the solution, the added quantity of AgNO_3 during titration, preferentially reacts with chloride ions and precipitates them as AgCl . At the end point, when chloride ions are completely precipitated as AgCl , further addition of AgNO_3 reacts with chromate ions forming brick red colored Ag_2CrO_4 .



- Thus the endpoint is indicated by appearance of red tinge.

- If the water sample is acidic, it has to be neutralized by addition of CaCO_3 . Otherwise, the presence of acid converts chromate ion of indicator to dichromate ion.



° observations:

Burette Level	Trial 1	Trial 2
Initial	0	0
Final	4.8	4.8
Volume of AgNO_3 (mL)	4.8	4.8

— Mean volume of $\text{AgNO}_3 = \underline{4.8 \text{ mL}}$

° Calculations:

— Volume of AgNO_3 used (V_1) = 4.8 mL

— Normality of AgNO_3 solution (N_1) = 0.05 N

— Volume of water sample (V_2) = 25 mL

— Normality of water sample,

$$N_2 = \frac{N_1 V_1}{V_2} = \frac{(0.05)(4.8)}{(25)} = 9.6 \times 10^{-3} \text{ N}$$

— Weight of chloride ions per litre solution

$$(W) = N_2 \times \text{Equivalent weight of } \text{Cl}^-$$

$$\therefore W = (9.6 \times 10^{-3}) \times (35.5) = \underline{0.3408 \text{ gm}}$$

— $\therefore$ Chloride content of solution = 0.3408 gm/L = 340.8 mg/L

- Procedure:
 - Pipette out 25 mL of water sample into a clean conical flask.
 - Add 5-6 drops of K_2CrO_4 indicator.
 - Titrate against $AgNO_3$ solution taken into the burette.
 - The endpoint is a brick red tinge.
- Result:
 - Chloride content in given sample of water is 340.8 mg/L.